BobSim

TransientEval Lateral Transient Response

Step response and sustained-sine FRF

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Run Notes

- Representative step steer response
- Continuous sinusoidal responses
- Rise time uses 50 percent input to 90 percent response thresholds
- Frequency delays are reported at 0.5 Hz and 1.0 Hz
- Gain and phase slopes summarize the FRF trend
- Steering shown in degrees for readability

TransientEval Metrics Summary 15.0 m/s

Time Domain - Step Response

STEP RESPONSE

a_y Rise Time (50-90%)	0.08	s
Sideslip Rise Time (50-90%)	0.19	s
Yaw Rise Time (50-90%)	0.06	s
Yaw Peak Response Time	0.10	s
a_y DC Gain	33.74	$\frac{m/s^2}{rad}$
Sideslip DC Gain	0.031	rad/rad
Yaw DC Gain	2.23	(rad/s)/rad
Roll DC Gain	0.071	rad/rad

STEADY-STATE / OVERSHOOT

a_y Peak	3.45	m/s^2
a_y Steady-State	2.84	m/s^2
a_y Overshoot	21.6	%
Yaw Steady-State	0.190	rad/s
Yaw Overshoot	0.035	rad/s
Roll Steady-State	0.006	rad
Roll Overshoot	0.001	rad
Settling Time	1.30	s

TransientEval Metrics Summary 15.0 m/s

Frequency Domain - Sustained Sine

FREQUENCY RESPONSE - CORE

DC Gain (a_y)	33.74	$\frac{m/s^2}{rad}$
DC Gain (r)	2.23	(rad/s)/rad
Peak Gain (a_y)	34.96	$\frac{m/s^2}{rad}$
Peak Freq (a_y)	1.00	Hz
Bandwidth (-3 dB)	1.00	Hz
Gain Slope (a_y)	1.01	dB/dec
Gain Slope (r)	0.28	dB/dec
Phase Slope (a_y)	11.48	deg/dec
Phase Slope (r)	16.59	deg/dec

DELAY / COUPLING

Ay/Yaw Delay @ 0.5 Hz	0.015	s
Yaw/Steer Delay @ 0.5 Hz	0.027	s
Difference @ 0.5 Hz	-0.013	s
Ay/Yaw Delay @ 1.0 Hz	0.017	s
Yaw/Steer Delay @ 1.0 Hz	0.028	s
Difference @ 1.0 Hz	-0.011	s
Gain Variation	3.6	%
Fit Error (a_y)	3.24e-03	-
Fit Error (r)	1.05e-03	-

TransientEval Metrics Summary 20.0 m/s

Time Domain - Step Response

STEP RESPONSE

a_y Rise Time (50-90%)	0.10	s
Sideslip Rise Time (50-90%)	0.09	s
Yaw Rise Time (50-90%)	0.06	s
Yaw Peak Response Time	0.11	s
a_y DC Gain	53.80	$\frac{m/s^2}{rad}$
Sideslip DC Gain	-0.018	rad/rad
Yaw DC Gain	2.69	(rad/s)/rad
Roll DC Gain	0.113	rad/rad

STEADY-STATE / OVERSHOOT

a_y Peak	4.96	m/s^2
a_y Steady-State	4.52	m/s^2
a_y Overshoot	9.8	%
Yaw Steady-State	0.227	rad/s
Yaw Overshoot	0.030	rad/s
Roll Steady-State	0.010	rad
Roll Overshoot	0.001	rad
Settling Time	1.23	s

TransientEval Metrics Summary 20.0 m/s

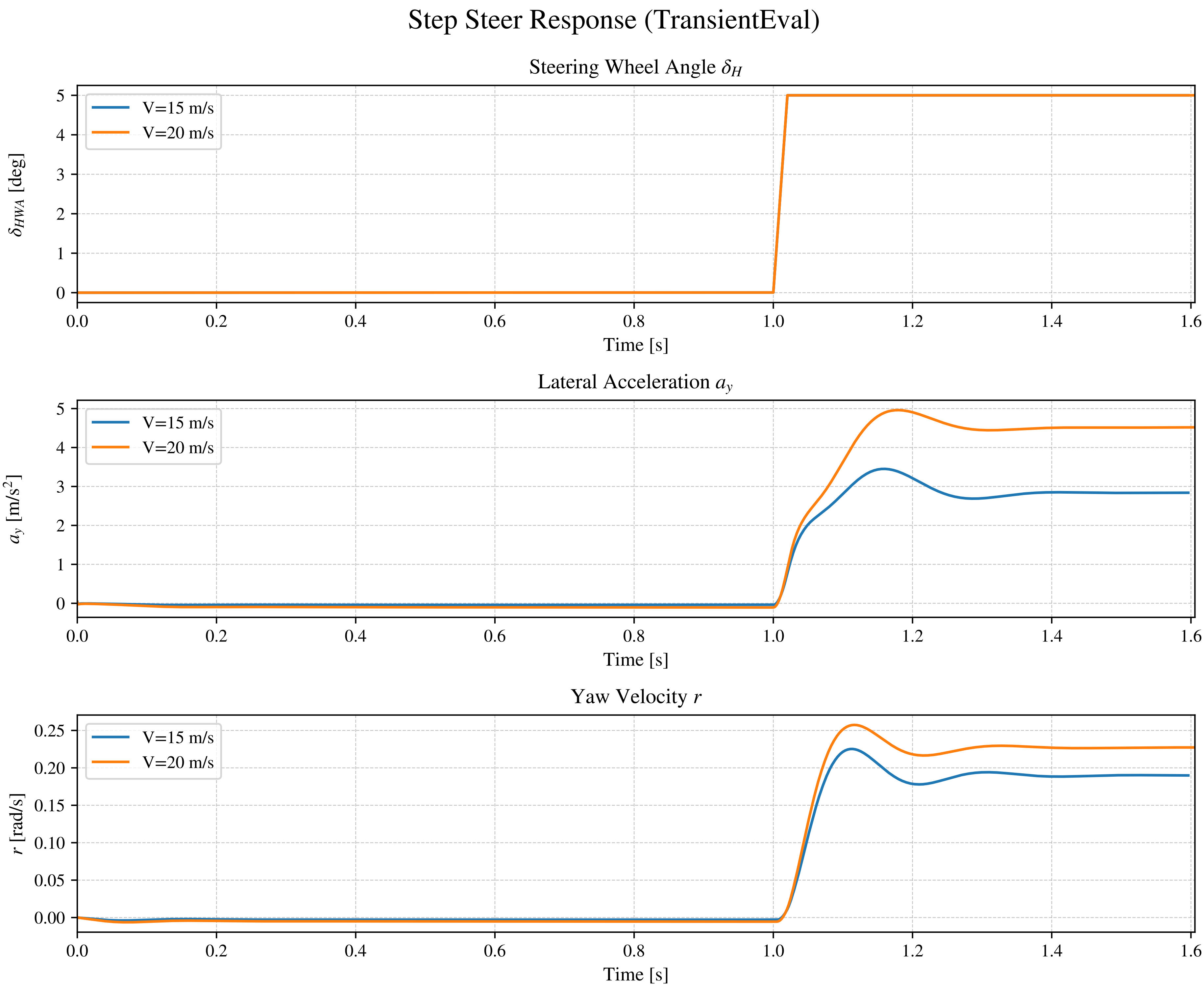
Frequency Domain - Sustained Sine

FREQUENCY RESPONSE - CORE

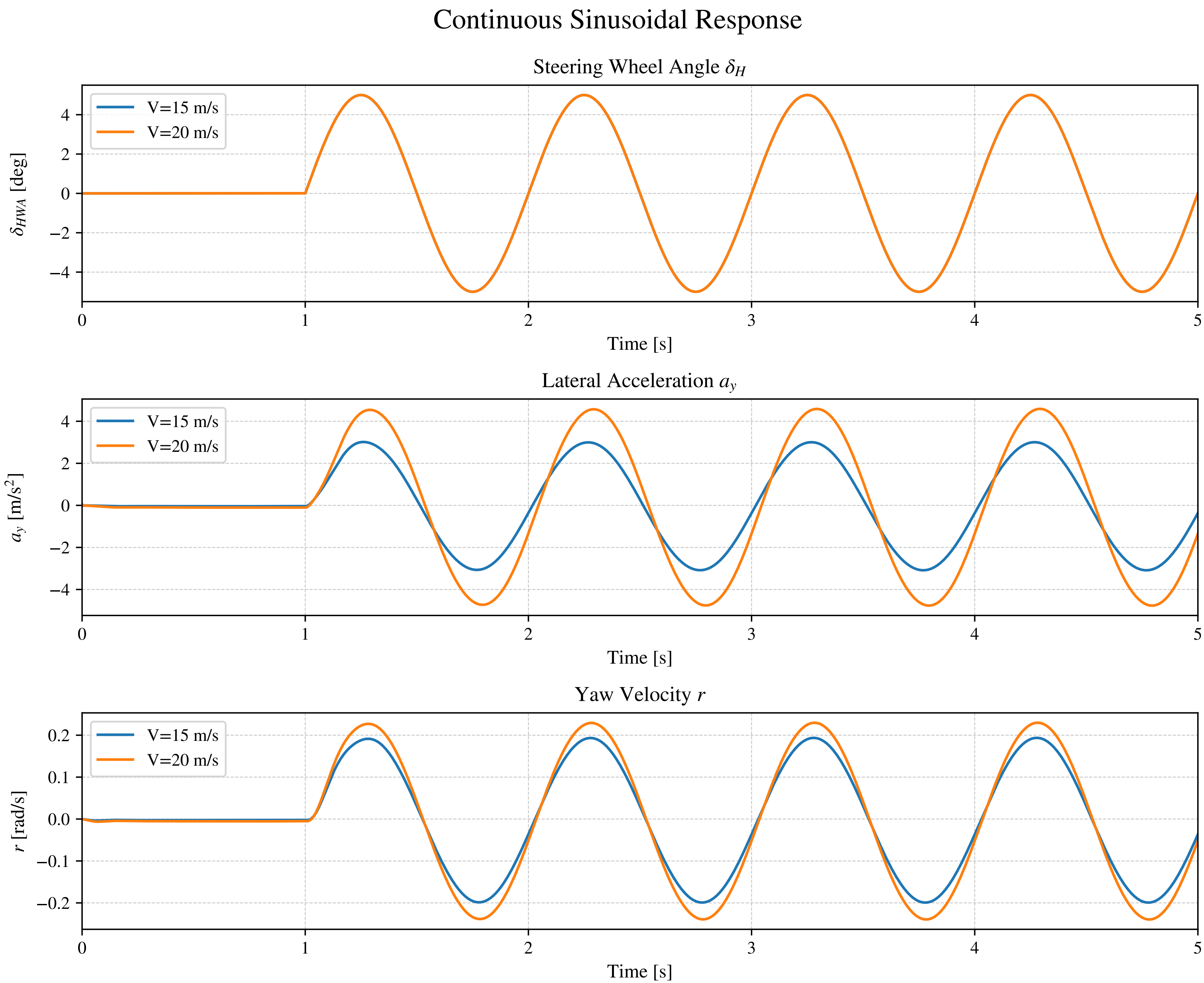
DC Gain (a_y)	53.80	$\frac{m/s^2}{rad}$
DC Gain (r)	2.69	(rad/s)/rad
Peak Gain (a_y)	53.89	$\frac{m/s^2}{rad}$
Peak Freq (a_y)	1.00	Hz
Bandwidth (-3 dB)	1.00	Hz
Gain Slope (a_y)	0.05	dB/dec
Gain Slope (r)	0.09	dB/dec
Phase Slope (a_y)	25.33	deg/dec
Phase Slope (r)	18.64	deg/dec

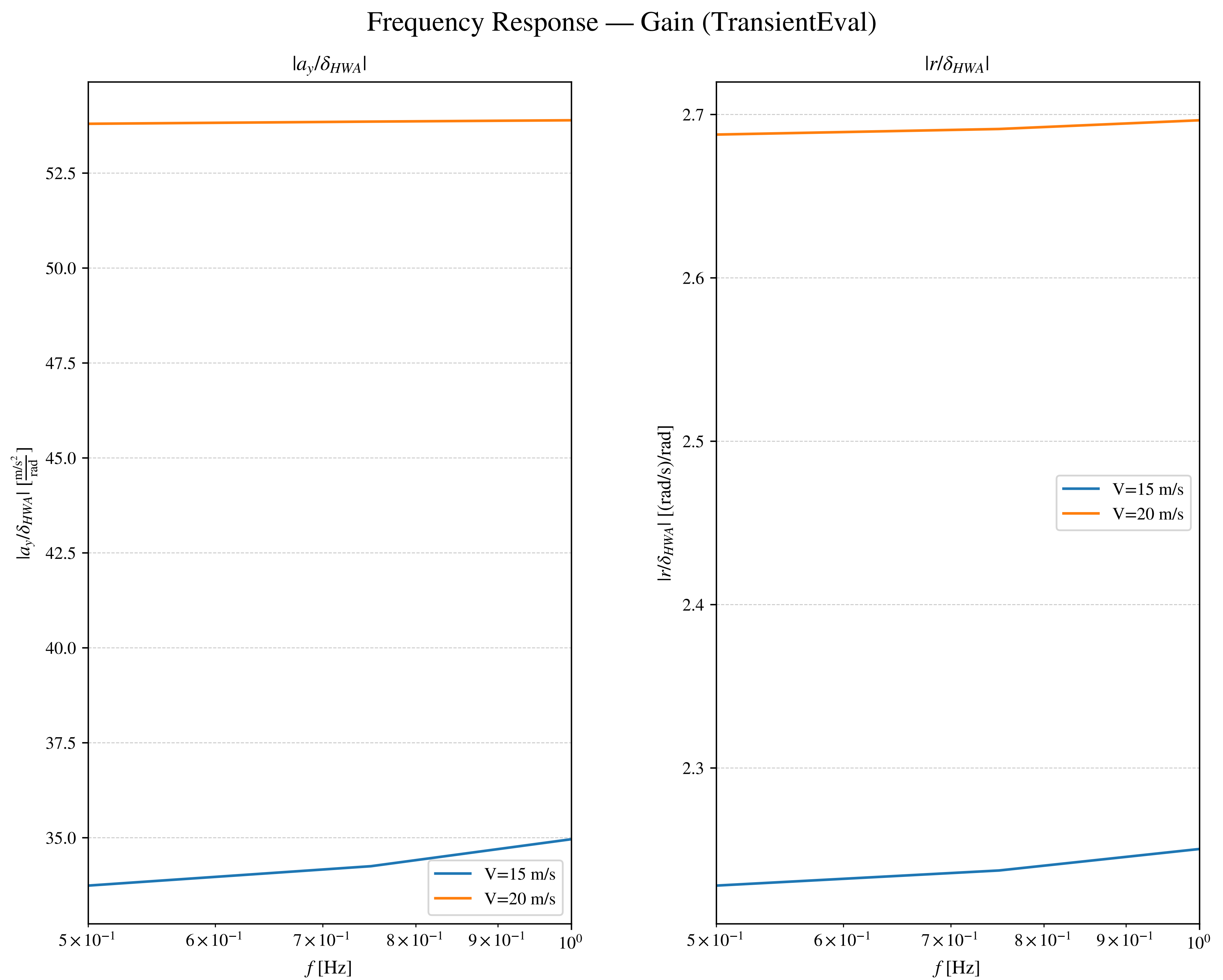
DELAY / COUPLING

Ay/Yaw Delay @ 0.5 Hz	0.040	s
Yaw/Steer Delay @ 0.5 Hz	0.032	s
Difference @ 0.5 Hz	0.009	s
Ay/Yaw Delay @ 1.0 Hz	0.042	s
Yaw/Steer Delay @ 1.0 Hz	0.032	s
Difference @ 1.0 Hz	0.010	s
Gain Variation	0.2	%
Fit Error (a_y)	7.19e-03	-
Fit Error (r)	4.19e-03	-



Step Steer Response (TransientEval)
Steering Wheel Angle δ_H
δ_{HWA} [deg]
V=15 m/s
V=20 m/s
Time [s]
Lateral Acceleration a_y
a_y [m/s^2]
V=15 m/s
V=20 m/s
Time [s]
Yaw Velocity r
r [rad/s]
V=15 m/s
V=20 m/s
Time [s]


Continuous Sinusoidal Response
Steering Wheel Angle δ_H
δ_{HWA} [deg]
V=15 m/s
V=20 m/s
4
2
0
−2
−4
0
1
2
3
4
5
Time [s]
Lateral Acceleration a_y
a_y [m/s^2]
V=15 m/s
V=20 m/s
4
2
0
−2
−4
0
1
2
3
4
5
Time [s]
Yaw Velocity r
r [rad/s]
V=15 m/s
V=20 m/s
0.2
0.1
0.0
−0.1
−0.2
0
1
2
3
4
5
Time [s]



Frequency Response — Gain (TransientEval)
$|a_y/\delta_{HWA}|$
$|a_y/\delta_{HWA}|$ $[\frac{m/s^2}{rad}]$
52.5
50.0
47.5
45.0
42.5
40.0
37.5
35.0
V=15 m/s
V=20 m/s
5×10^{-1}
6×10^{-1}
7×10^{-1}
8×10^{-1}
9×10^{-1}
10^0
f [Hz]
$|r/\delta_{HWA}|$
$|r/\delta_{HWA}|$ [(rad/s)/rad]
2.7
2.6
2.5
2.4
2.3
V=15 m/s
V=20 m/s
5×10^{-1}
6×10^{-1}
7×10^{-1}
8×10^{-1}
9×10^{-1}
10^0
f [Hz]

Frequency Response — Phase (TransientEval)